

EMo Learners

PROFESSIONAL QUESTION BANK

Analysis and Design of Algorithms (ADA)

Unit 1 & Unit 2 Objective Questions

Subject: Analysis and Design of Algorithms

Total Questions: 24

Scope: Unit 1 (Complexity Analysis, Sorting, Searching, Divide-and-Conquer) & Unit 2 (Greedy Approach)

Document Status: Verified Question Bank

Unit 1 & Unit 2: Core Algorithm Design & Analysis MCQs

1. What is the worst-case time for merge sort to sort an array of n elements?

- A. $O(\log n)$
- B. $O(n)$
- C. $O(n \log n)$
- D. $O(n^2)$

Answer: C. $O(n \log n)$

Solution: Merge sort consistently divides the array in half and merges them, taking $O(n \log n)$ time in all cases.

2. What is the worst-case time for quick sort to sort an array of n elements?

- A. $O(\log n)$
- B. $O(n)$
- C. $O(n \log n)$
- D. $O(n^2)$

Answer: D. $O(n^2)$

Solution: This occurs when the array is already sorted and the smallest or largest element is consistently chosen as the pivot.

3. In a selection sort of n elements, how many times is the swap function called in the complete execution of the algorithm?

- A. 1
- B. $n - 1$
- C. $n \log n$
- D. n^2

Answer: B. $n - 1$

Solution: Selection sort makes exactly one swap for each position in the array as it builds the sorted portion.

4. Selection sort and quick sort both fall into the same category of sorting algorithms. What is this category?

- A. $O(n \log n)$ sorts
- B. Divide-and-conquer sorts
- C. Interchange sorts
- D. Average time is quadratic

Answer: C. Interchange sorts

Solution: Both algorithms fundamentally rely on swapping or interchanging elements based on comparisons to arrange them.

5. What is the Best-case time for Binary Search to sort an array of n elements? (Note: Binary Search is a searching algorithm)

- A. $O(\log n)$
- B. $O(1)$
- C. $O(n \log n)$
- D. $O(n^2)$

Answer: B. $O(1)$

Solution: The best case occurs when the middle element of the array is the exact target element on the very first comparison.

6. Consider the following array - 23, 32, 45, 69, 72, 73, 89, 97. Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?

- A. Selection Sort
- B. Merge Sort
- C. Insertion Sort
- D. Quick Sort

Answer: C. Insertion Sort

Solution: The given array is already sorted. Insertion sort takes $O(n)$ comparisons for an already sorted array, which is the minimum among the options.

7. The Master Theorem (Main Recurrence Theorem)

- A. is used to prove correctness of algorithms
- B. gives general solutions to time recurrences for divide-and-conquer algorithms
- C. gives intractability results
- D. is proven by temporal logic

Answer: B. gives general solutions to time recurrences for divide-and-conquer algorithms

Solution: The Master Theorem provides a cookbook method for solving recurrences of the form $T(n) = aT(n/b) + f(n)$ arising in divide-and-conquer algorithms.

8. Knapsack is a(n) ____ problem.

- A. non-optimization vector, matrix or graph
- B. optimization
- C. state-space search
- D. behavior-of-program

Answer: B. optimization

Solution: The goal is to maximize profit/value while keeping the total weight under a specific capacity.

9. The binary search uses which approach to algorithm design?

- A. divide and conquer
- B. greedy
- C. brute force
- D. dynamic programming

Answer: A. divide and conquer

Solution: It repeatedly divides the search interval in half.

10. Graph path search involves finding a:

- A. set of vertices
- B. sequence of vertices
- C. set of edges
- D. minimal set of edges

Answer: B. sequence of vertices

Solution: A path in a graph is defined as an ordered sequence of vertices connected by edges.

11. What is the time complexity of Huffman Coding?

- A. $O(\log n)$
- B. $O(n)$
- C. $O(n \log n)$
- D. $O(n^2)$

Answer: C. $O(n \log n)$

Solution: Building the initial min-heap takes $O(n)$, and extracting the two minimum elements $n - 1$ times takes $O(n \log n)$.

12. Consider a job scheduling problem with 4 jobs J_1, J_2, J_3, J_4 and with corresponding deadlines: $(d_1, d_2, d_3, d_4) = (4, 2, 4, 2)$. Which of the following is not a feasible schedule without violating any job schedule?

- A. J_3, J_2, J_1, J_4
- B. J_4, J_1, J_2, J_3
- C. J_4, J_2, J_1, J_3
- D. J_2, J_4, J_1, J_3

Answer: A. J_3, J_2, J_1, J_4

Solution: In option A, Job J_4 is executed in the 4th time slot, but its deadline is 2. Therefore, it violates the deadline constraint.

13. Which asymptotic notation is used to represent the lower bound of an algorithm's running time?

- A. Big-Oh (O)
- B. Big-Omega (Ω)
- C. Theta (Θ)
- D. Little-oh (o)

Answer: B. Big-Omega (Ω)

Solution: It provides the minimum time an algorithm will take for large inputs (asymptotically tight lower bound).

14. What is the time complexity of Strassen's Matrix Multiplication algorithm?

- A. $O(n^3)$
- B. $O(n^2)$
- C. $O(n^{2.81})$
- D. $O(n \log n)$

Answer: C. $O(n^{2.81})$

Solution: Strassen's algorithm reduces the 8 recursive multiplications to 7, leading to a recurrence of $T(n) = 7T(n/2) + O(n^2)$, which solves to $O(n^{\log_2 7}) \approx O(n^{2.81})$.

15. Which of the following algorithmic strategies is best suited to solve the Fractional Knapsack problem optimally?

- A. Dynamic Programming
- B. Greedy Strategy
- C. Divide and Conquer
- D. Backtracking

Answer: B. Greedy Strategy

Solution: Taking items sorted by the highest profit-to-weight ratio (value density) yields the optimal solution for the fractional version.

16. Which data structure is primarily utilized in Kruskal's algorithm to efficiently check for cycles in a graph?

- A. Min-Heap
- B. Queue
- C. Disjoint Set (Union-Find)
- D. Stack

Answer: C. Disjoint Set (Union-Find)

Solution: It keeps track of the connected components and quickly verifies if adding an edge connects two already connected vertices.

17. What is the recurrence relation for the standard Merge Sort algorithm?

- A. $T(n) = T(n/2) + O(1)$
- B. $T(n) = 2T(n/2) + O(1)$
- C. $T(n) = 2T(n/2) + O(n)$
- D. $T(n) = T(n - 1) + O(n)$

Answer: C. $T(n) = 2T(n/2) + O(n)$

Solution: The array is split into two halves of size $n/2$, and merging them takes linear $O(n)$ time.

18. Which of the following sorting algorithms is considered "stable"?

- A. Quick Sort
- B. Heap Sort
- C. Selection Sort
- D. Merge Sort

Answer: D. Merge Sort

Solution: A stable sort maintains the relative order of equal elements. Merge sort preserves this order during the merge phase.

19. Dijkstra's Algorithm is used to solve which of the following problems?

- A. Minimum Spanning Tree
- B. Single-Source Shortest Path
- C. All-Pairs Shortest Path
- D. Maximum Flow

Answer: B. Single-Source Shortest Path

Solution: It finds the shortest paths from one specific source vertex to all other vertices in a graph with non-negative edge weights.

20. If an algorithm has a time complexity of $\Theta(n^2)$, this implies that:

- A. Its worst-case is $O(n^2)$ but best case could be $O(n)$.
- B. Its running time grows exactly quadratically for all cases.
- C. Its running time is bounded both above and below by a quadratic function.
- D. It is faster than an $O(n \log n)$ algorithm.

Answer: C. Its running time is bounded both above and below by a quadratic function.

Solution: Theta notation (Θ) signifies an asymptotically tight bound, meaning the upper and lower bounds grow at the same rate.

21. In Prim's Algorithm for finding a Minimum Spanning Tree, which data structure is optimal for selecting the next edge with the minimum weight?

- A. Priority Queue (Min-Heap)
- B. Stack
- C. Hash Table
- D. Linked List

Answer: A. Priority Queue (Min-Heap)

Solution: It efficiently allows extraction of the minimum edge weight connecting the current tree to unvisited vertices in $O(\log V)$ time.

22. According to the Master Theorem, if $T(n) = aT(n/b) + f(n)$ and $f(n) = O(n^c)$ where $c < \log_b a$, then the time complexity is:

- A. $\Theta(f(n))$
- B. $\Theta(n^{\log_b a})$
- C. $\Theta(n^c \log n)$
- D. $\Theta(n \log n)$

Answer: B. $\Theta(n^{\log_b a})$

Solution: This is Case 1 of the Master Theorem, where the work at the leaves of the recursion tree dominates the total work.

23. What type of tree is constructed during the generation of Huffman Codes?

- A. Binary Search Tree
- B. AVL Tree
- C. Full Binary Prefix Tree
- D. B-Tree

Answer: C. Full Binary Prefix Tree

Solution: Huffman coding builds a binary tree from the bottom up where all characters are leaves, ensuring no code is a prefix of another.

24. The optimal binary merge pattern problem can be efficiently solved using which approach?

- A. Divide and Conquer
- B. Greedy Strategy
- C. Dynamic Programming
- D. Backtracking

Answer: B. Greedy Strategy

Solution: By always choosing to merge the two smallest files first, usually implemented with a min-heap, we achieve the optimal merging cost.